## 11.4 Continuous Distributions

In this section we learn

(i) Continuous random variable

(ii) Probability density function

(iii) Distribution function (Cumulative distribution function).

(iv) To determine distribution function from probability density function.

(v) To determine probability density function from distribution function.

Sometimes a measurement such as current in a copper wire or length of lifetime of an electric bulb, can assume any value in an interval of real numbers. Then any precision in the measurement is possible. The random variable that represents this measurement is said to be a **continuous** random variable. The range of the random variable includes all values in an interval of real numbers; that is, the range can be thought of as a continuum of real numbers

### 11.4.1 The definition of continuous random variable

**Definition 11.5 (Continuous Random Variable)**

*Let $S$ be a sample space and let a random variable $X : S \to R$ that takes on any value in a set $I$ of $R$. Then $X$ is called a **continuous random variable** if $P(X = x) = 0$ for every $x$ in $I$*

### 11.4.2 Probability density function

**Definition 11.6: (Probability density function)**

*A non-negative real valued function $f(x)$ is said to be a **probability density function** if, for each possible outcome x, $x \in [a, b]$ of a continuous random variable X having the property*

$$P(a \le X \le b) = \int_a^b f(x)\, dx$$

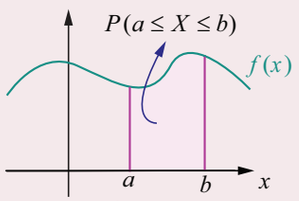


Fig. 11.11

**Theorem 11.2 (Without proof)**

A function $f(.)$ is a probability density function for some continuous random variable $X$ if and only if it satisfies the following properties.

(i) $f(x) \ge 0$, for every $x$ and (ii) $\int_{-\infty}^{\infty} f(x)dx = 1$ .

**Note**

It follows from the above definition, if $X$ is a continuous random variable,

$$P(a \le X \le b) = \int_a^b f(x)\, dx, \text{ which means that } P(X = a) = \int_a^a f(x)\, dx = 0$$

That is probability when $X$ takes on any one particular value is zero.



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